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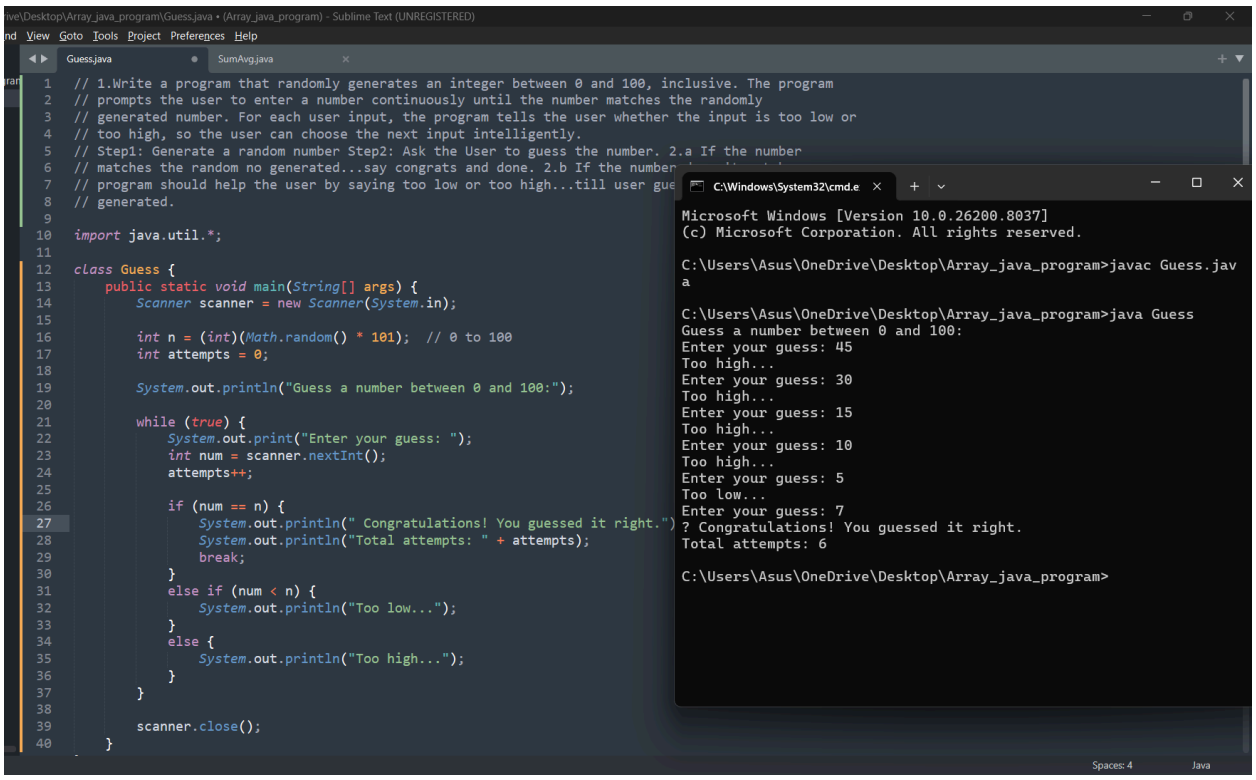
Batch: PG-DAC February 2026
Subject: JAVA
Assignment: 4
Form Number: 251101107

// 1.Write a program that randomly generates an integer between 0 and 100, inclusive. The program
// prompts the user to enter a number continuously until the number matches the randomly
// generated number. For each user input, the program tells the user whether the input is too low or
// too high, so the user can choose the next input intelligently.
// Step1: Generate a random number Step2: Ask the User to guess the number. 2.a If the number
// matches the random no generated...say congrats and done. 2.b If the number doesn't match,
// program should help the user by saying too low or too high...till user guesses the random number
// generated.

Solution:

```
import java.util.*;

class Guess {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int n = (int)(Math.random() * 101);  // 0 to 100
        int attempts = 0;

        System.out.println("Guess a number between 0 and 100:");

        while (true) {
            System.out.print("Enter your guess: ");
            int num = scanner.nextInt();
            attempts++;

            if (num == n) {
                System.out.println(" Congratulations! You guessed it right.");
                System.out.println("Total attempts: " + attempts);
                break;
            }
```

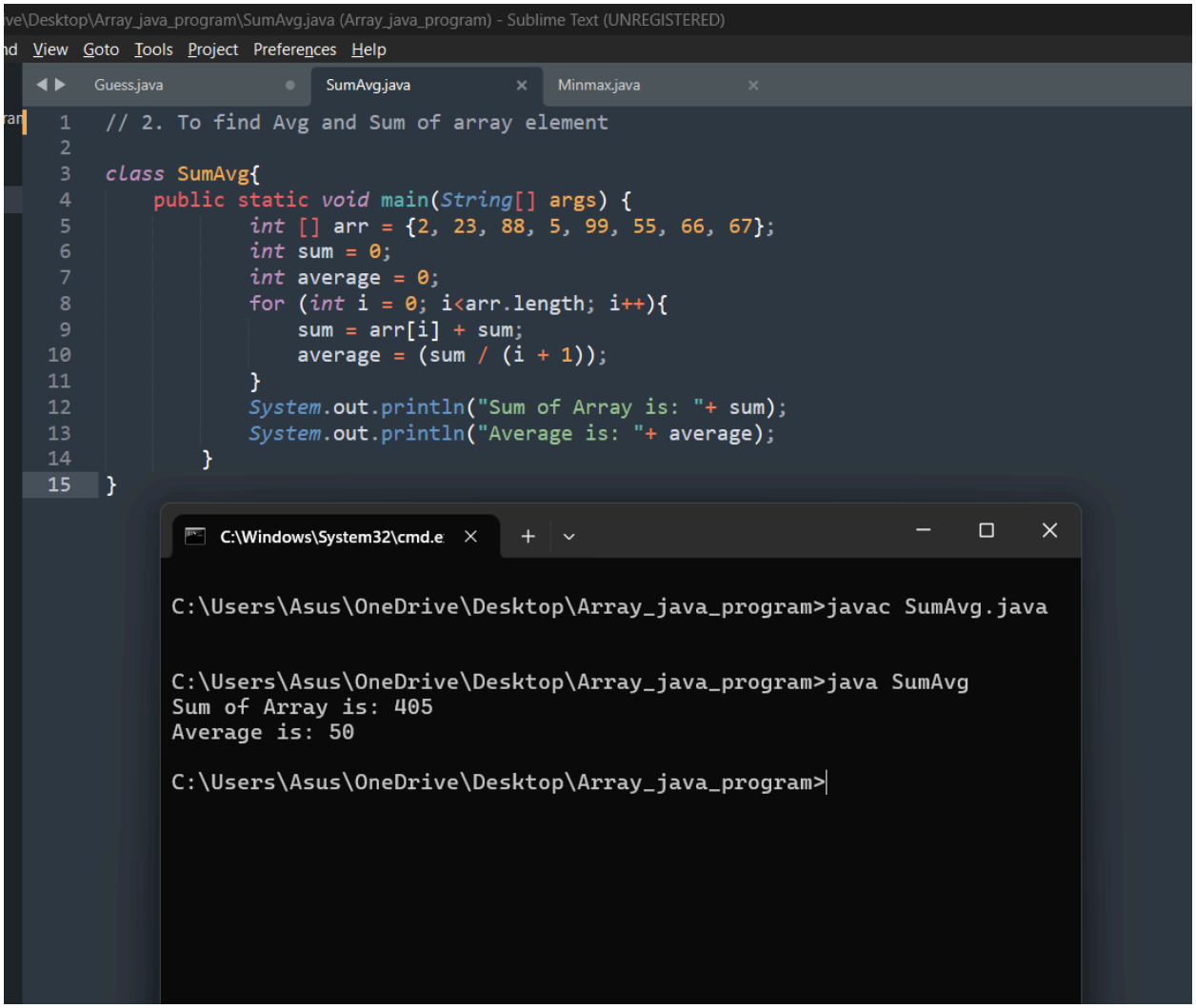
```
        else if (num < n) {
            System.out.println("Too low...");
        }
        else {
            System.out.println("Too high...");
        }
    }
}
}
```



// 2. To find Avg and Sum of array element
Solution:

```
class SumAvg{
        public static void main(String[] args) {
                        int [] arr = {2, 23, 88, 5, 99, 55, 66, 67};
                        int sum = 0;
                        int average = 0;
                        for (int i = 0; i<arr.length; i++){
                                sum = arr[i] + sum;
                                average = (sum / (i + 1));
```

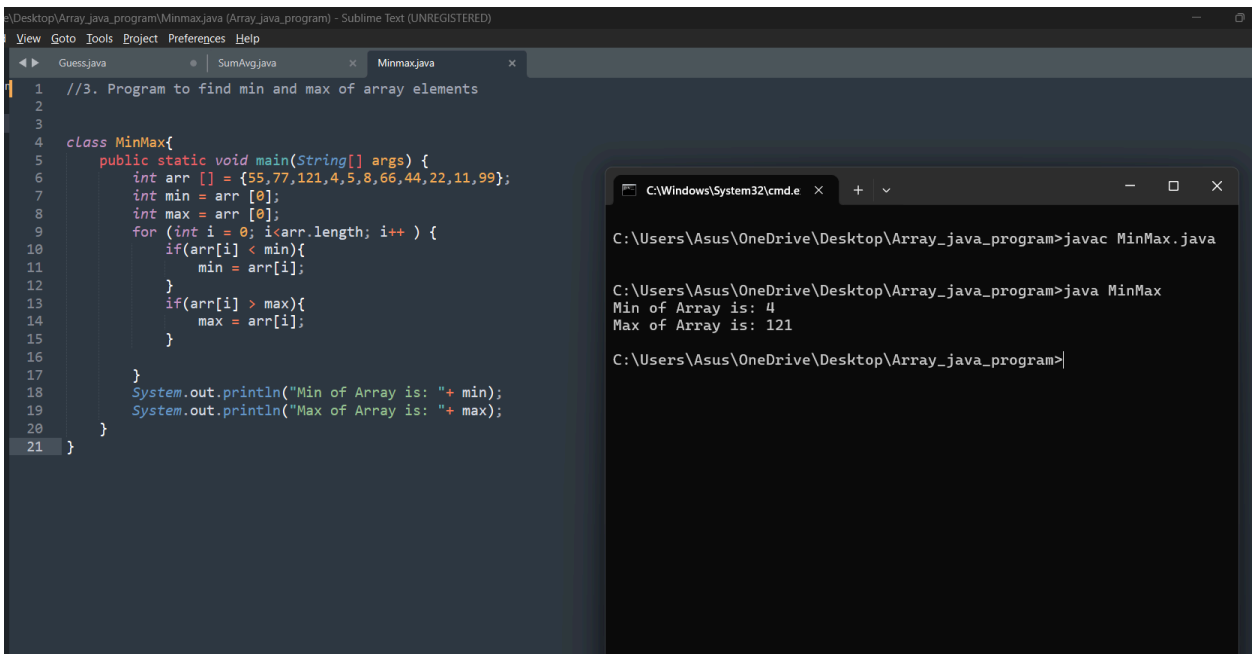
```
                        }
                        System.out.println("Sum of Array is: "+ sum);
                        System.out.println("Average is: "+ average);
                }
}
```



//3. Program to find min and max of array elements
Solution:

```
class MinMax{
        public static void main(String[] args) {
                int arr [] = {55,77,121,4,5,8,66,44,22,11,99};
                int min = arr [0];
                int max = arr [0];
                for (int i = 0; i<arr.length; i++ ) {
```

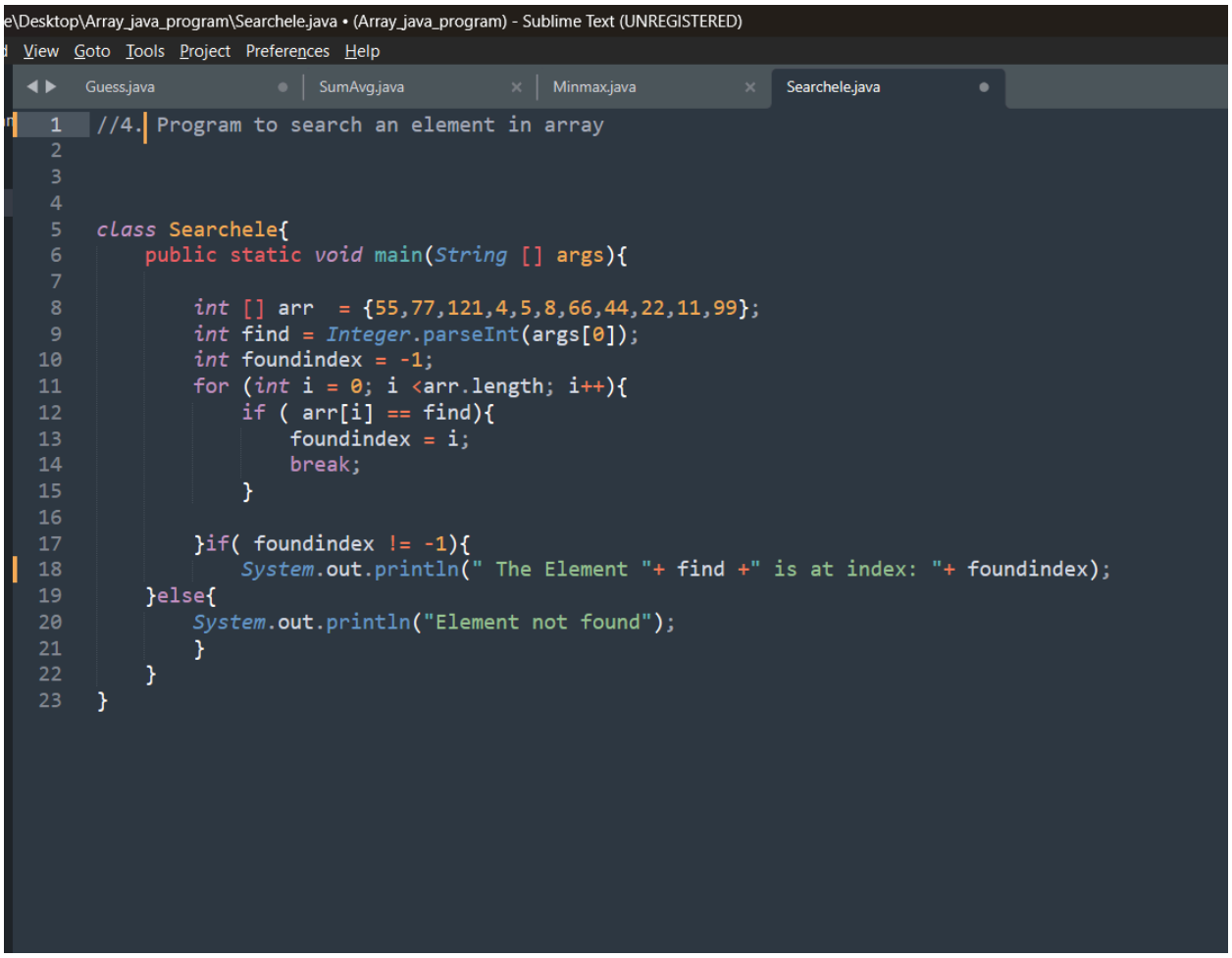
```
                if(arr[i] < min){
                        min = arr[i];
                }
                if(arr[i] > max){
                        max = arr[i];
                }

        }
        System.out.println("Min of Array is: "+ min);
        System.out.println("Max of Array is: "+ max);
    }
}
```



//4. Program to search an element in array
Solution:

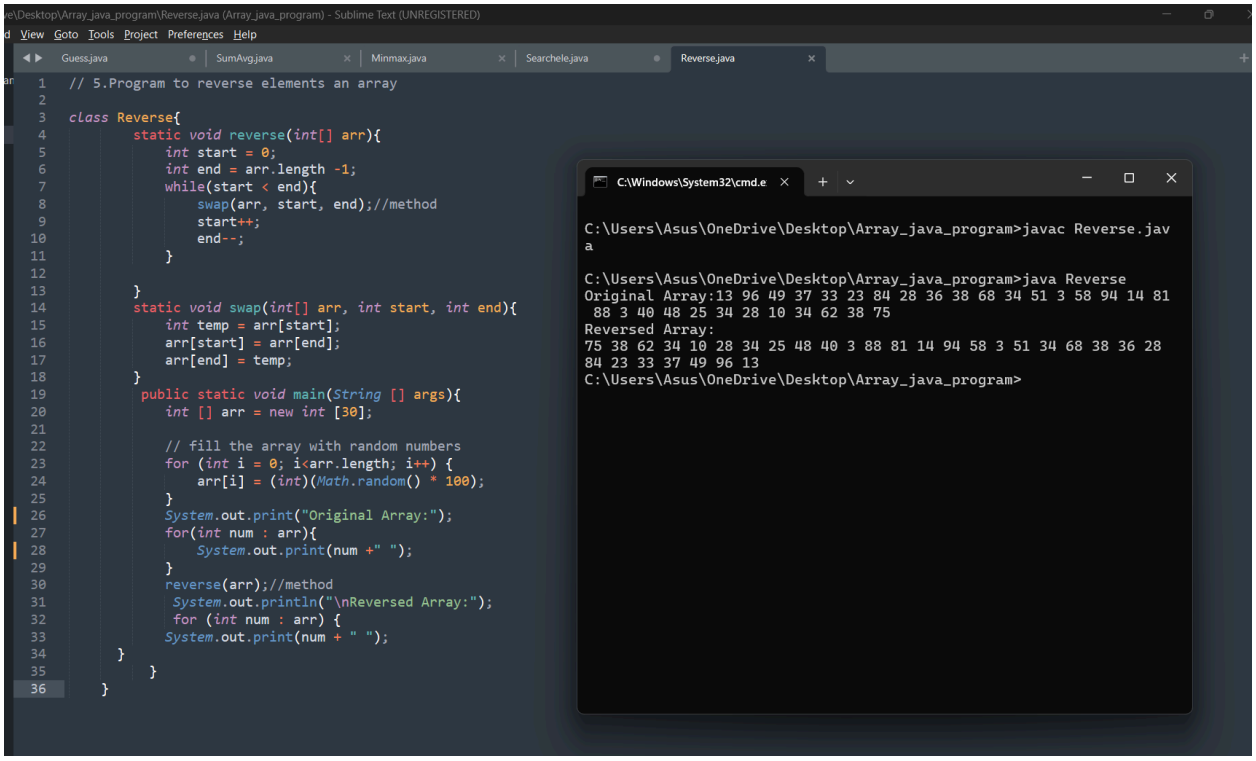
```
class Searchele{
        public static void main(String [] args){

                int [] arr  = {55,77,121,4,5,8,66,44,22,11,99};
                int find = Integer.parseInt(args[0]);
                int foundindex = -1;
                for (int i = 0; i <arr.length; i++){
```

```
                    if ( arr[i] == find){
                            foundindex = i;
                        break;
                    }

                }if( foundindex != -1){
                    System.out.println(" The Element "+ find +" is at index: "+ foundindex);
        }else{
                System.out.println("Element not found");
                }
        }
}
```



// 5.Program to reverse elements an array
Solution:

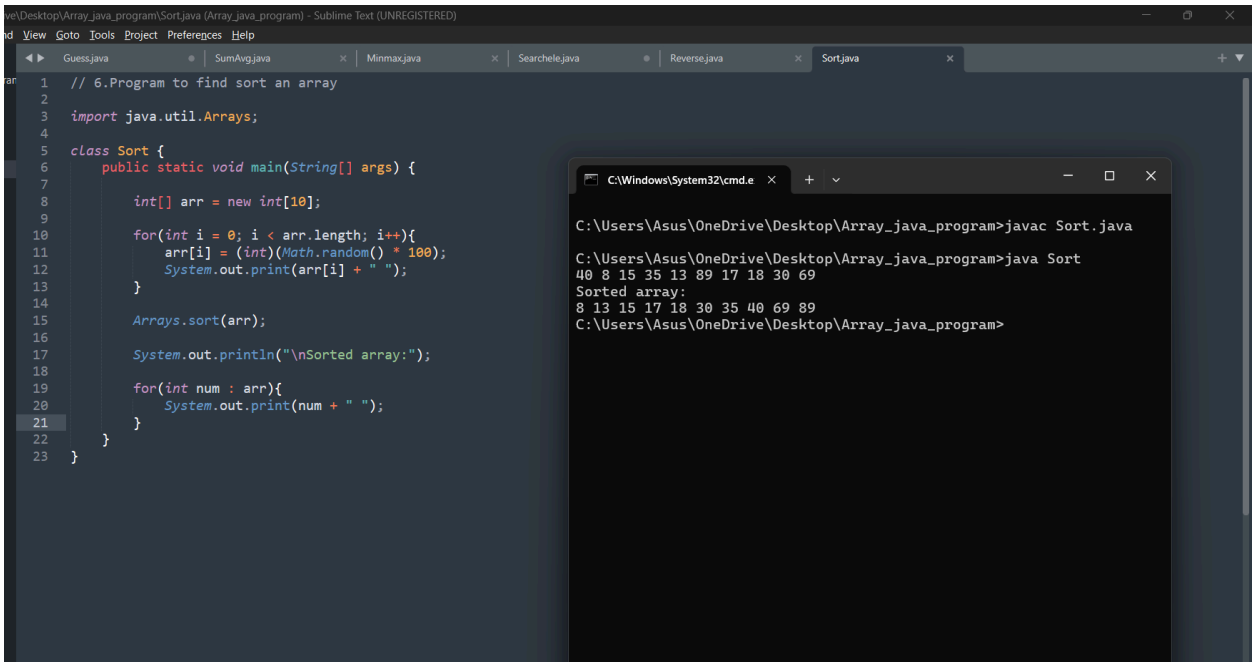
```
class Reverse{
		static void reverse(int[] arr){
			int start = 0;
			int end = arr.length -1;
			while(start < end){
				swap(arr, start, end);//method
				start++;
				end--;
			}

		}
		static void swap(int[] arr, int start, int end){
			int temp = arr[start];
			arr[start] = arr[end];
			arr[end] = temp;
		}
	public static void main(String [] args){
	  int [] arr = new int [30];

	  // fill the array with random numbers
	  for (int i = 0; i<arr.length; i++) {
		arr[i] = (int)(Math.random() * 100);
	  }
	  System.out.println("Original Array:");
	  for(int num : arr){
		System.out.println(num +" ");
	  }
	  reverse(arr);//method
	   System.out.println("\nReversed Array:");
		 for (int num : arr) {
	  System.out.print(num + " ");
    }
	   }
	  }
```



// 6.Program to find sort an array
Solution:

```
import java.util.Arrays;

class Sort {
   public static void main(String[] args) {

      int[] arr = new int[10];

      for(int i = 0; i < arr.length; i++){
         arr[i] = (int)(Math.random() * 100);
         System.out.print(arr[i] + " ");
      }

      Arrays.sort(arr);

      System.out.println("\nSorted array:");

      for(int num : arr){
         System.out.print(num + " ");
```

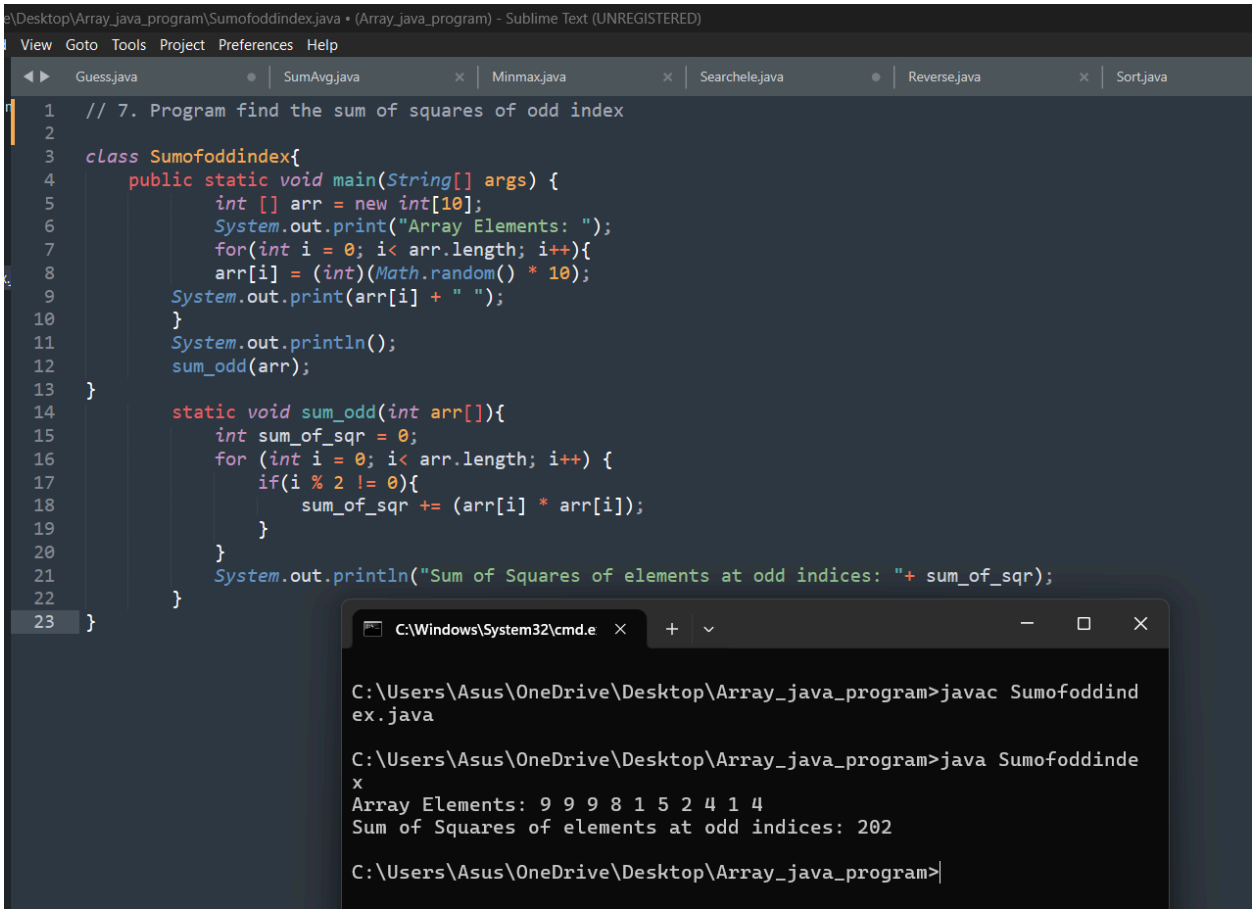
```
        }
    }
}
```



// 7. Program find the sum of squares of odd index
Solution:

```
class Sumofoddindex{
        public static void main(String[] args) {
                int [] arr = new int[10];
                System.out.print("Array Elements: ");
                for(int i = 0; i< arr.length; i++){
                arr[i] = (int)(Math.random() * 10);
            System.out.print(arr[i] + " ");
            }
            System.out.println();
            sum_odd(arr);
}
            static void sum_odd(int arr[]){
                int sum_of_sqr = 0;
                for (int i = 0; i< arr.length; i++) {
                    if(i % 2 != 0){
                        sum_of_sqr += (arr[i] * arr[i]);
                    }
```

```
                        }
                        System.out.println("Sum of Squares of elements at odd indices: "+
sum_of_sqr);
                }
}
```

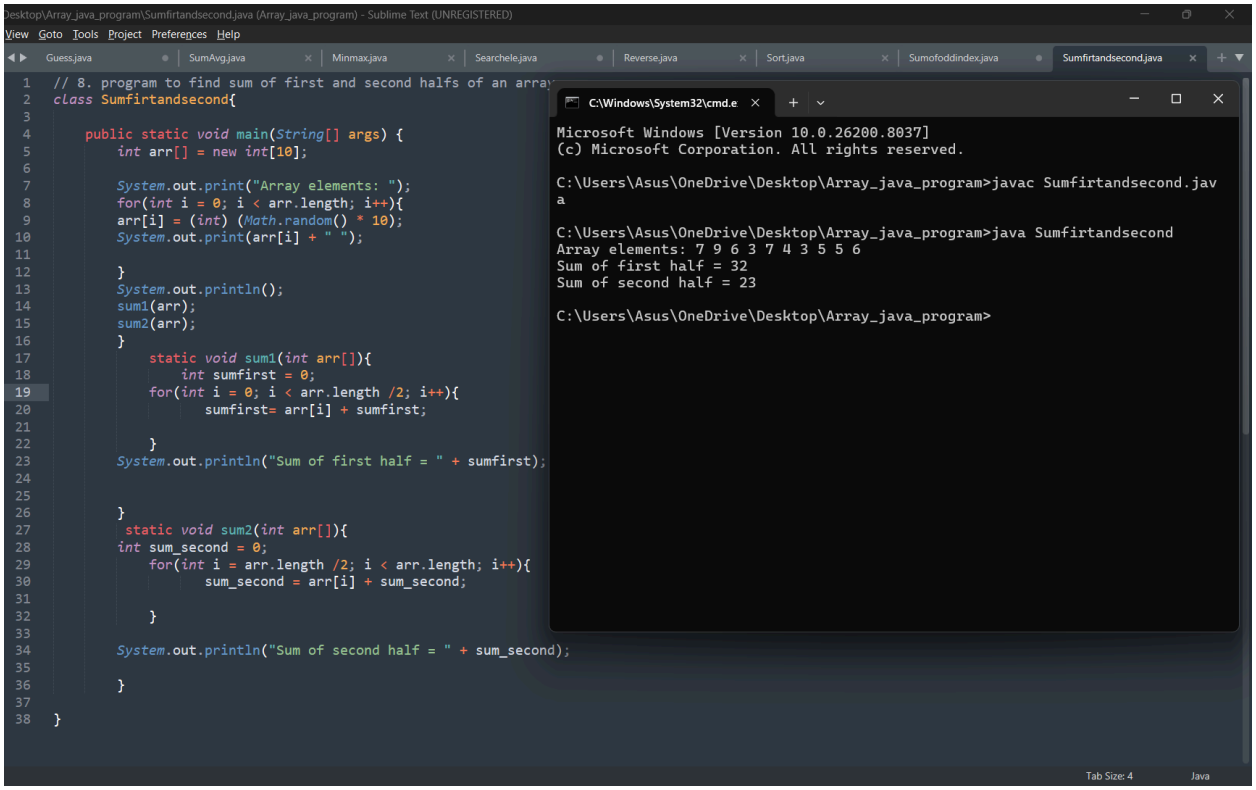


// 8. program to find sum of first and second halfs of an array.
class SumfirsndSecd{

Solution:

```
        public static void main(String[] args) {
                int arr[] = new int[10];

                System.out.print("Array elements: ");
                for(int i = 0; i < arr.length; i++){
```

```
            arr[i] = (int) (Math.random() * 10);
            System.out.print(arr[i] + " ");

            }
            System.out.println();
            sum1(arr);
            sum2(arr);
            }
                    static void sum1(int arr[]){
                            int sumfirst = 0;
                    for(int i = 0; i < arr.length /2; i++){
                                sumfirst= arr[i] + sumfirst;

                    }
    System.out.println("Sum of first half = " + sumfirst);


            }
        static void sum2(int arr[]){
            int sum_second = 0;
                    for(int i = arr.length /2; i < arr.length; i++){
                                sum_second = arr[i] + sum_second;

                    }

    System.out.println("Sum of second half = " + sum_second);

            }

}
```

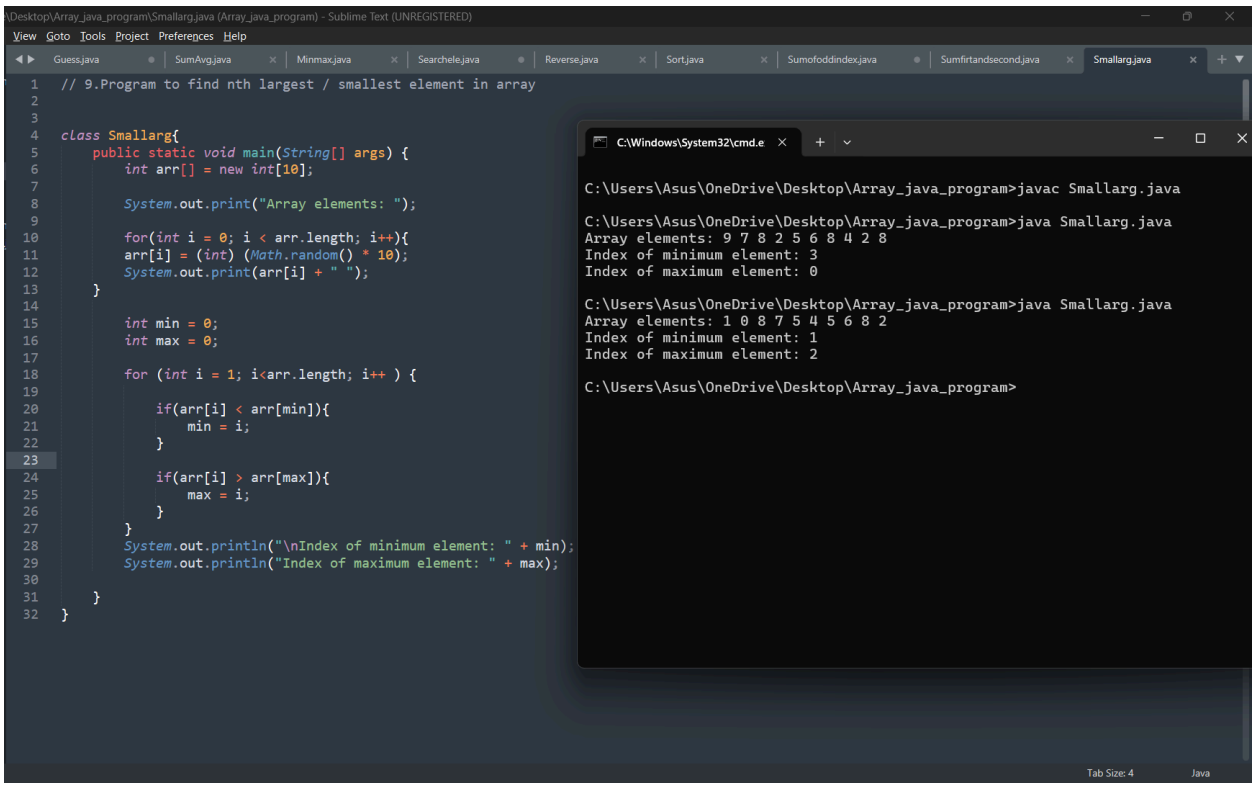


// 9.Program to find nth largest / smallest element in array

Solution:

```
class Smallarg{
        public static void main(String[] args) {
                int arr[] = new int[10];

                System.out.print("Array elements: ");

                for(int i = 0; i < arr.length; i++){
                arr[i] = (int) (Math.random() * 10);
                System.out.print(arr[i] + " ");
        }

                int min = 0;
                int max = 0;

                for (int i = 1; i<arr.length; i++ ) {

                        if(arr[i] < arr[min]){
                                min = i;
                        }
```

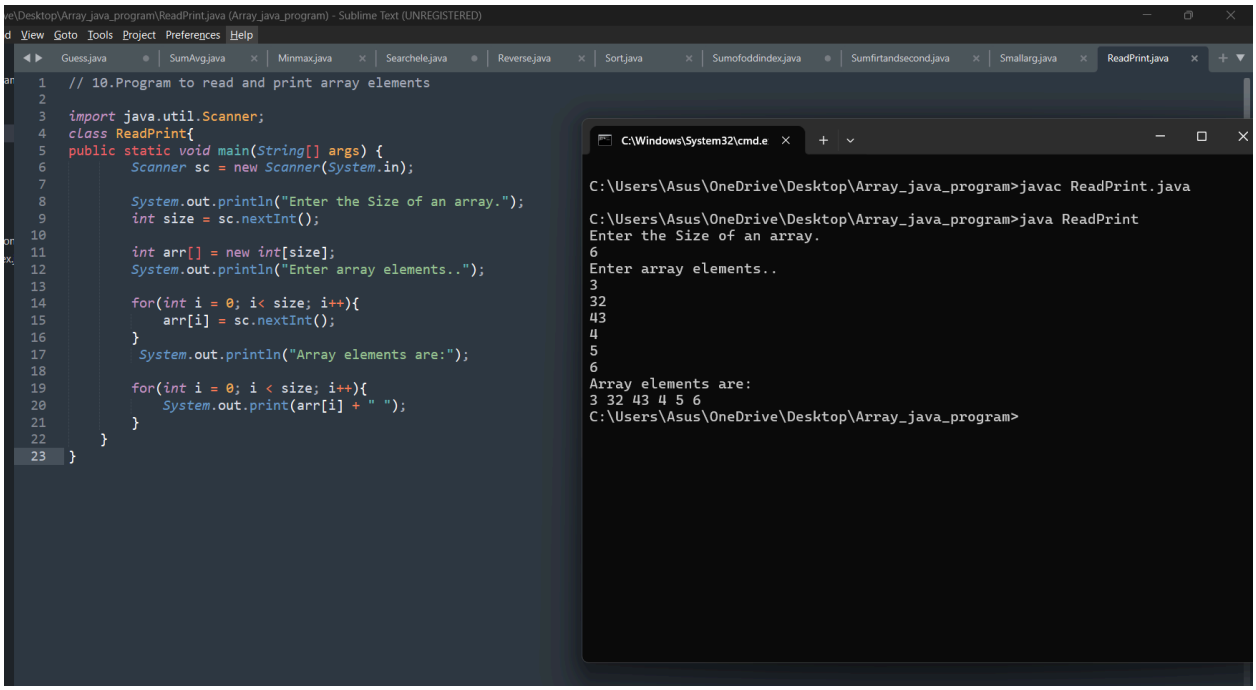
```
                if(arr[i] > arr[max]){
                        max = i;
                }
        }
        System.out.println("\nIndex of minimum element: " + min);
        System.out.println("Index of maximum element: " + max);

    }
}
```



// 10.Program to read and print array elements
Solution:

```
import java.util.Scanner;
class ReadPrint{
public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter the Size of an array.");
```

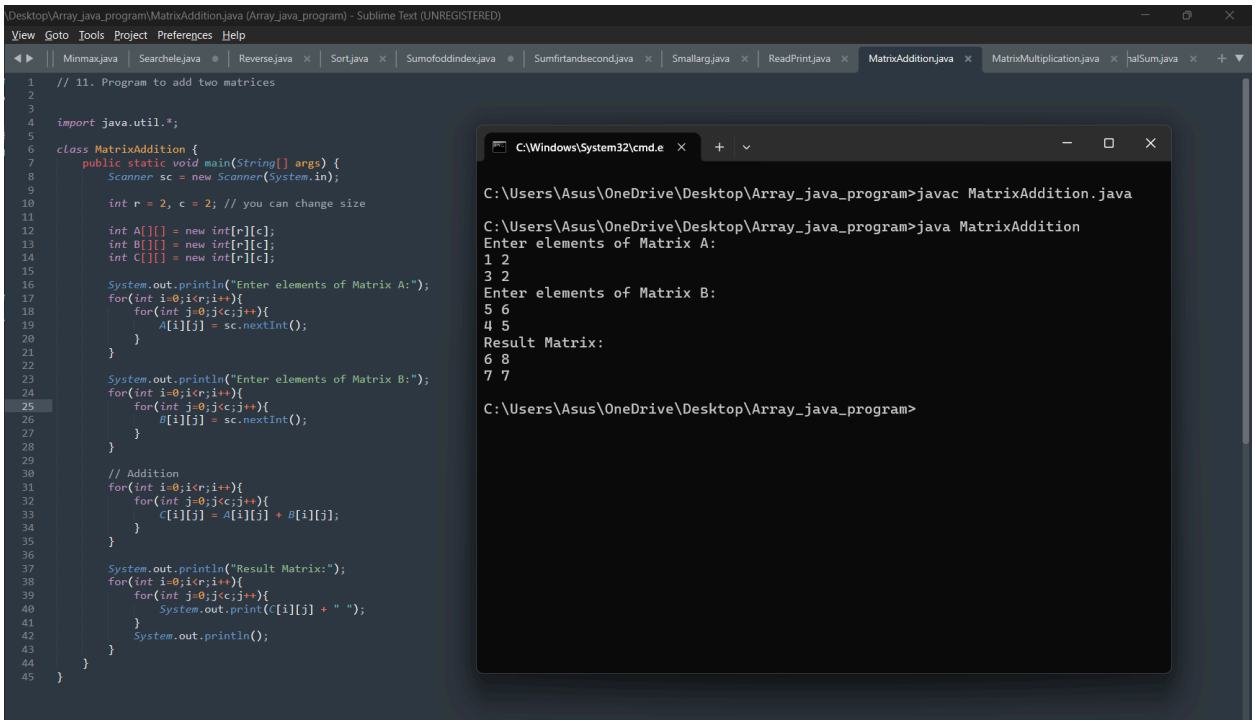
```
            int size = sc.nextInt();

            int arr[] = new int[size];
    System.out.println("Enter array elements..");

            for(int i = 0; i< size; i++){
                    arr[i] = sc.nextInt();
            }
             System.out.println("Array elements are:");

    for(int i = 0; i < size; i++){
       System.out.print(arr[i] + " ");
    }
       }
}
```



// 11. Program to add two matrices
Solution:

```
import java.util.*;

class MatrixAddition {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int r = 2, c = 2; // you can change size

        int A[][] = new int[r][c];
        int B[][] = new int[r][c];
        int C[][] = new int[r][c];

        System.out.println("Enter elements of Matrix A:");
        for(int i=0;i<r;i++){
            for(int j=0;j<c;j++){
                A[i][j] = sc.nextInt();
            }
        }

        System.out.println("Enter elements of Matrix B:");
        for(int i=0;i<r;i++){
            for(int j=0;j<c;j++){
                B[i][j] = sc.nextInt();
            }
        }

        // Addition
        for(int i=0;i<r;i++){
            for(int j=0;j<c;j++){
                C[i][j] = A[i][j] + B[i][j];
            }
        }

        System.out.println("Result Matrix:");
        for(int i=0;i<r;i++){
            for(int j=0;j<c;j++){
                System.out.print(C[i][j] + " ");
            }
            System.out.println();
        }
    }
```

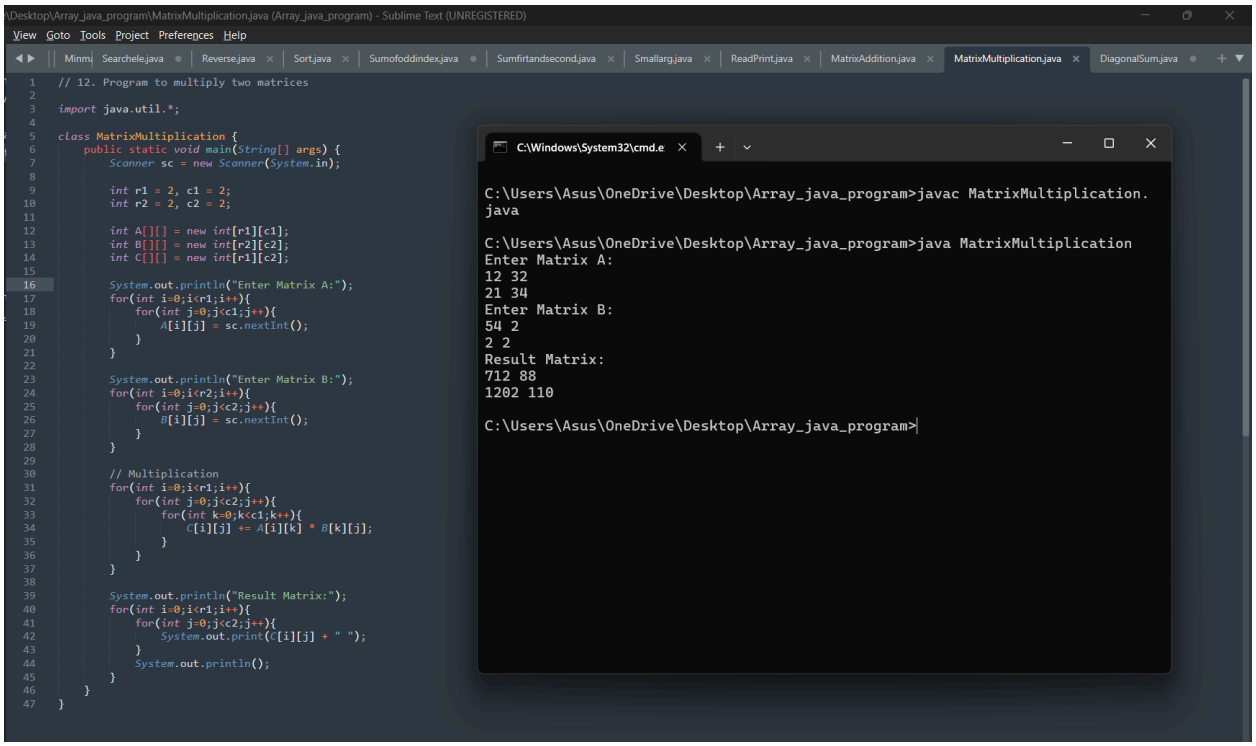
}



// 12. Program to multiply two matrices
Solution:

```
import java.util.*;

class MatrixMultiplication {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int r1 = 2, c1 = 2;
      int r2 = 2, c2 = 2;

      int A[][] = new int[r1][c1];
      int B[][] = new int[r2][c2];
      int C[][] = new int[r1][c2];

      System.out.println("Enter Matrix A:");
      for(int i=0;i<r1;i++){
         for(int j=0;j<c1;j++){
```

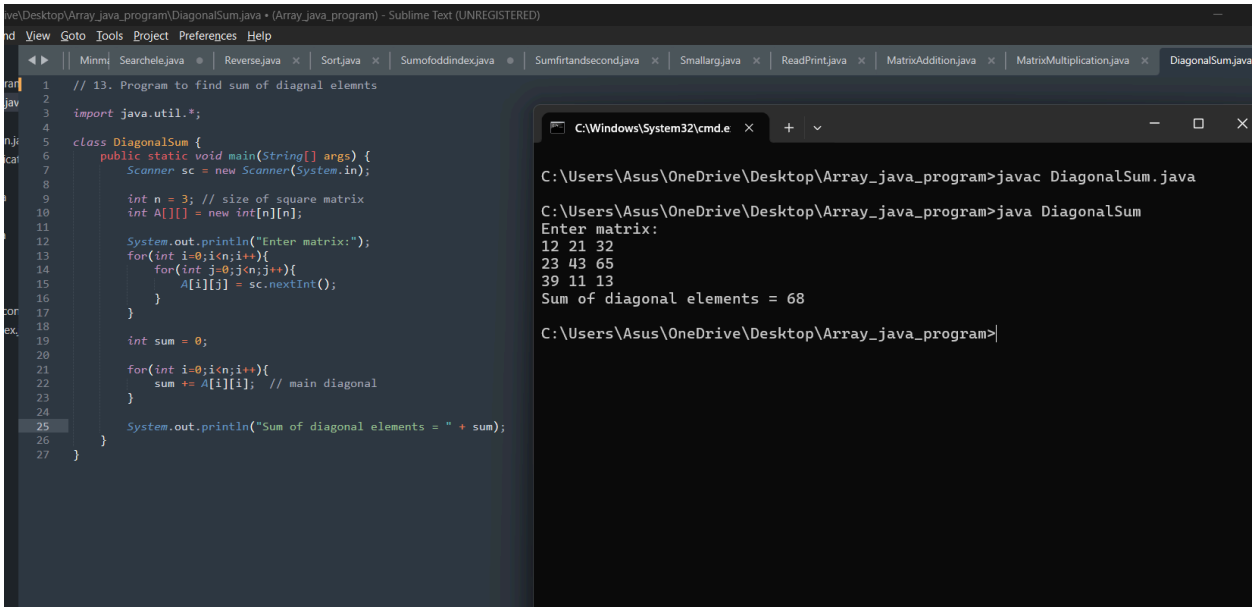
```
            A[i][j] = sc.nextInt();
        }
    }

    System.out.println("Enter Matrix B:");
    for(int i=0;i<r2;i++){
        for(int j=0;j<c2;j++){
            B[i][j] = sc.nextInt();
        }
    }

    // Multiplication
    for(int i=0;i<r1;i++){
        for(int j=0;j<c2;j++){
            for(int k=0;k<c1;k++){
                C[i][j] += A[i][k] * B[k][j];
            }
        }
    }

    System.out.println("Result Matrix:");
    for(int i=0;i<r1;i++){
        for(int j=0;j<c2;j++){
            System.out.print(C[i][j] + " ");
        }
        System.out.println();
    }
  }
}
```



// 13. Program to find sum of diagnal elemnts.
Solution:

```
import java.util.*;

class DiagonalSum {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int n = 3; // size of square matrix
      int A[][] = new int[n][n];

      System.out.println("Enter matrix:");
      for(int i=0;i<n;i++){
         for(int j=0;j<n;j++){
            A[i][j] = sc.nextInt();
         }
      }

      int sum = 0;

      for(int i=0;i<n;i++){
         sum += A[i][i];  // main diagonal
```

```
        }

        System.out.println("Sum of diagonal elements = " + sum);
    }
}
```



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